

B.Tech. Degree VI Semester Examination April 2019**ME 15-1601 DYNAMICS OF MACHINERY**
(2015 Scheme)

Time : 3 Hours

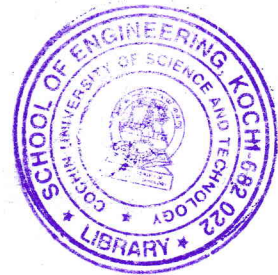
Maximum Marks : 60

PART A

(Answer ALL questions)

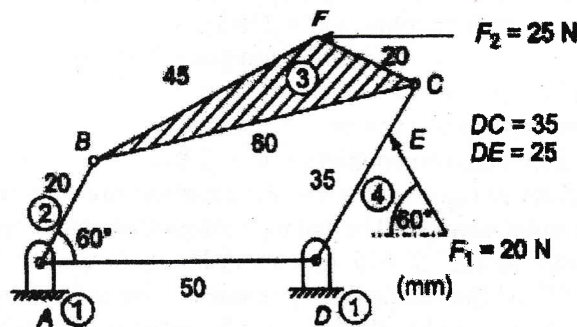
(10 × 2 = 20)

- I. (a) Differentiate between static force analysis and dynamic force analysis.
 (b) What do you mean by dynamically equivalent system?
 (c) State and explain D'Alembert's principle.
 (d) Explain the turning moment diagram of a multi-cylinder engine.
 (e) What is the condition for self-locking in brakes.
 (f) Explain the terms, spin and precession as related to gyroscopic effects. How do they differ from each other?
 (g) Why complete balancing is not possible in reciprocating masses?
 (h) What do you mean by static and dynamic balancing? What are the necessary condition to achieve them.
 (i) Distinguish between initial tension and centrifugal tension in belt.
 (j) What is the difference between a governor and a flywheel?

**PART B**

(4 × 10 = 40)

- II. For the mechanism shown in the figure, determine the torque on link AB for the static equilibrium of the mechanism. (10)



OR

- III. (a) The crank and connecting rod of a petrol engine, running at 1800 r.p.m. are 50 mm and 200 mm respectively. The diameter of the piston is 80 mm and the mass of the reciprocating parts is 1 kg. At a point during the power stroke, the pressure on the piston is 0.7 N/mm^2 , when it has moved 10 mm from the inner dead centre. Determine: (i) Net load on the gudgeon pin (ii) Thrust in the connecting rod (iii) Reaction between the piston and cylinder (iv) The engine speed at which the above values become zero. (6)
- (b) In a slider crank mechanism, the length of the crank and connecting rod are 150 mm and 600 mm respectively. The crank position is 60° from inner dead centre. The crank shaft speed is 450 r.p.m. (clockwise). Using analytical method, determine: (i) velocity and acceleration of the slider (ii) angular velocity and angular acceleration of the connecting rod. (4)
- IV. (a) The turning moment curve for an engine is represented by the equation, $T = (20000 + 9500 \sin 2\theta - 5700 \cos 2\theta) \text{ N-m}$, where θ is the angle moved by the crank from inner dead centre. If the resisting torque is constant, find: (i) Power developed by the engine (ii) Moment of inertia of flywheel in kg-m^2 , if the total fluctuation of speed is not to exceed 1% of mean speed which is 180 r.p.m. (iii) Angular acceleration of the flywheel when the crank has turned through 45° from inner dead centre. (6)

(P.T.O.)

- (b) Prove that the maximum fluctuation of energy (4)

$$\Delta E = E \times 2C_s$$

where E = Mean kinetic energy of the flywheel, and
 C_s = Coefficient of fluctuation of speed.

OR

- V. (a) Explain the effect of gyroscopic couple on ship. (5)

- (b) A band brake acts on the $3/4^{\text{th}}$ of circumference of a drum of 450 mm diameter which is keyed to the shaft. The band brake provides a braking torque of 225 N-m. One end of the band is attached to a fulcrum pin of the lever and the other end to pin 100 mm from the fulcrum. If the operating force is applied at 500 mm from the fulcrum and the coefficient of friction is 0.25, find the operating force when the drum rotates in the (i) anticlockwise direction (ii) clockwise direction. (5)

- VI. A shaft carries four masses in parallel planes A, B, C and D in this order along its length. The masses at B and C are 18 kg and 12.5 kg respectively, and each has an eccentricity of 60 mm. The masses at A and D have an eccentricity of 80 mm. The angle between the masses at B and C is 100° and that between the masses at B and A is 190° , both being measured in the same direction. The axial distance between the planes A and B is 100 mm and that between B and C is 200 mm. If the shaft is in complete dynamic balance, determine: (i) The magnitude of the masses at A and D (ii) the distance between planes A and D (iii) The angular position of the mass at D. (10)

OR

- VII. The following particulars relate to a two-cylinder locomotive with two coupled wheels on each side: (10)

Stroke = 650 mm

Mass of reciprocating parts per cylinder = 240 kg

Mass of revolving parts per cylinder = 200 kg

Mass of each coupling rod = 250 kg

Radius of centre of coupling rod pin = 250 mm

Distance between cylinders = 0.6 m

Distance between wheels = 1.5 m

Distance between coupling rods = 1.8 m

The main cranks are at right angles and the coupling rod pins are at 180° to their respective main cranks. The balance masses are to be placed in the wheels at a mean radius of 675 mm in order to balance whole of the revolving and $3/4^{\text{th}}$ of the reciprocating masses. The balance mass for the reciprocating masses is to be divided equally between the driving wheels and the coupled wheels. Find: (i) The magnitudes and angular positions of the masses required for the driving and trailing wheels and (ii) The hammer blow at 120 km/hr if the wheels are 1.8 metre diameter.

- VIII. (a) A leather belt is required to transmit 7.5 kW from a pulley 1.2 m in diameter, running at 250 r.p.m. The angle embraced is 165° and the coefficient of friction between the belt and the pulley is 0.3. If the safe working stress for the leather belt is 1.5 MPa, density of leather 1000 kg/m^3 and thickness of belt 10 mm, determine the width of the belt taking centrifugal tension into account. (6)

- (b) Derive the condition for the transmission of the maximum power in a flat belt drive. (4)

OR

- IX. (a) What is stability of a governor? Sketch the controlling force versus radius diagrams for a stable, unstable and isochronous governor. Derive the conditions for stability. (4)

- (b) A Proell governor has equal arms of length 300 mm. The upper and lower ends of the arms are pivoted on the axis of the governor. The extension arms of the lower links are each 80 mm long and parallel to the axis when the radii of rotation of the balls are 150 mm and 200 mm. The mass of each ball is 10 kg and the mass of the central load is 100 kg. Determine the range of speed of the governor. (6)